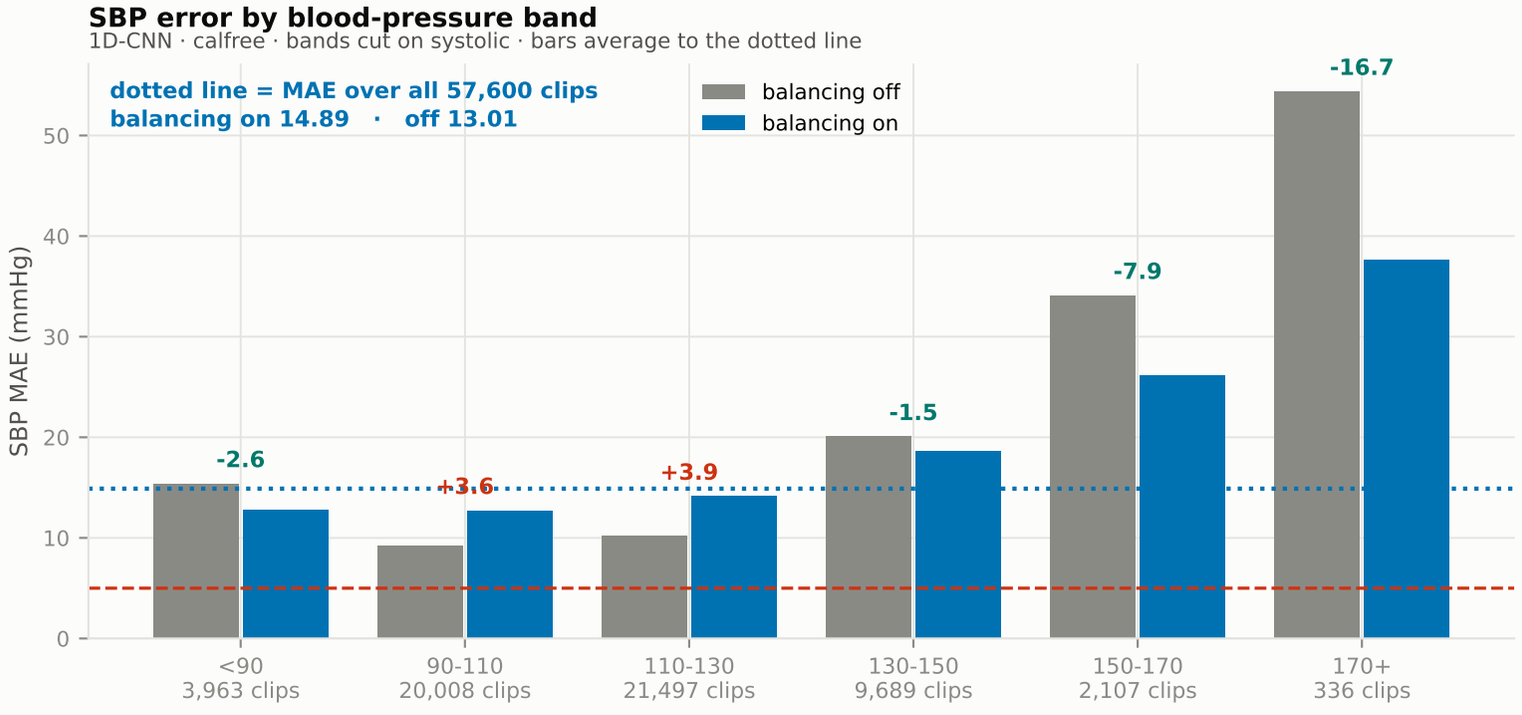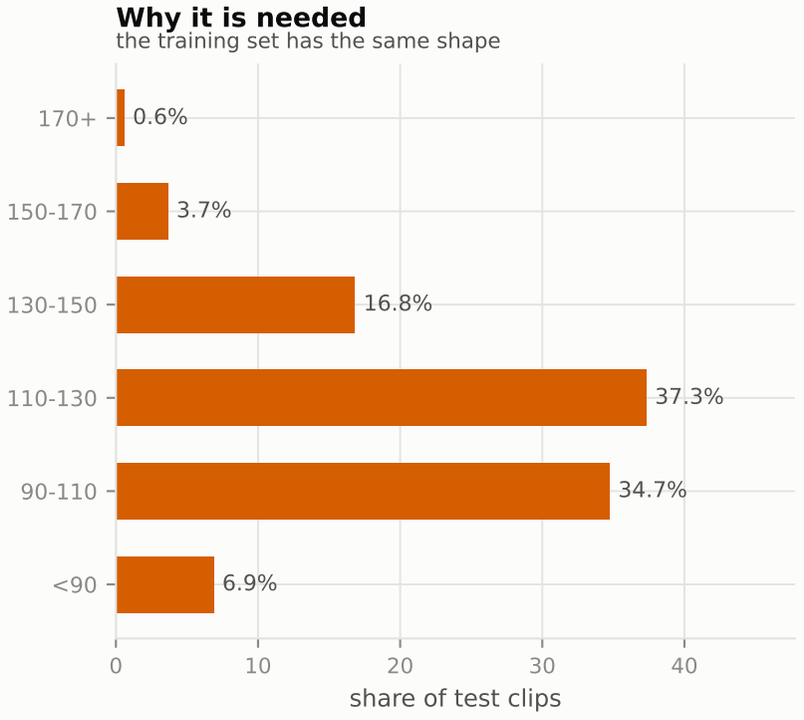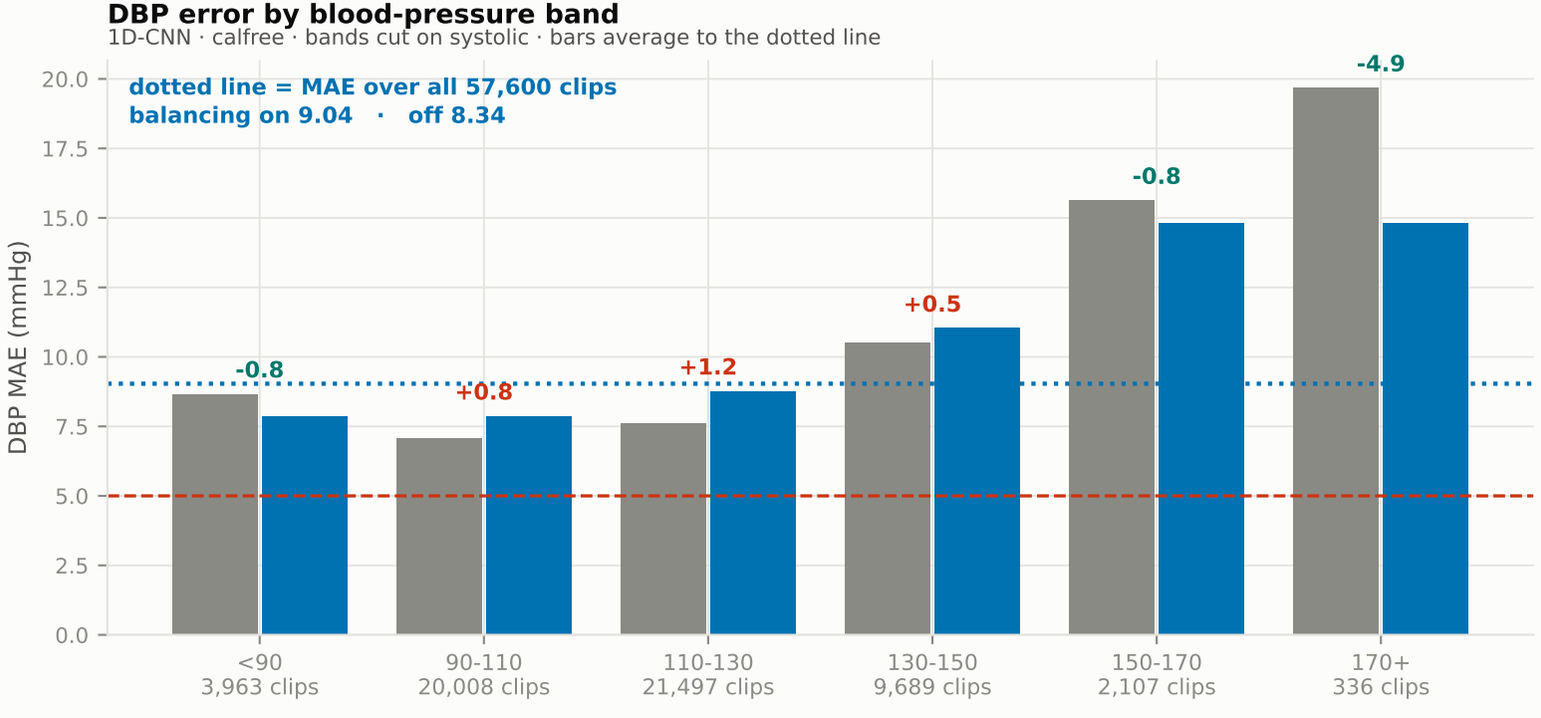Balancing costs a little on the average and buys a lot on the patients who matter — 1D-CNN